# Session 4: Cleaning day

*Question: What does the data look like when it is clean, and how much did we sell?*

## Concepts

| Concept | In daily life | In Excel | The limit |
|---|---|---|---|
| DataFrame, read_csv | The whole warehouse list | A sheet, File > Open | The path must be exact |
| head, shape, info | First page, page count, cover | Scrolling, COUNTA | Looking first is part of the job |
| rename with a dict | Relabelling bins | Editing headers | Uses the dict from Session 2 |
| filter | Keeping only the good boxes | Filter | Save the result, or it is lost. Use & with brackets |
| new column | Writing a value on each box | A formula column | No auto-update. Run it again |
| date and month | A calendar column | A date cell | Convert to a real date first |
| .copy() | Save As with a new name | Save As | Changing the copy cannot hurt the original |
| NaN | An empty box on a form | An empty cell | isna() counts them |

## Code patterns

```
sales = pd.read_csv("path/to/file.csv")
sales.head()
print(sales.shape)

sales_clean = sales_clean[sales_clean["quantity"] > 0]
sales_clean = sales_clean[~sales_clean["invoice"].str.startswith("C")]

both = sales_clean[(sales_clean["quantity"] > 0) & (sales_clean["price"] > 0)]

sales_clean["revenue"] = sales_clean["quantity"] * sales_clean["price"]
sales_clean["month"] = sales_clean["invoice_date"].dt.to_period("M")
```

## Three common errors

| Error | What to do |
|---|---|
| FileNotFoundError | The path is wrong. Copy the path from the Drive folder and check the spelling. |
| KeyError after rename | You used the old column name. Use the new short names. |
| `and` instead of `&` | Use & between conditions and put each condition in brackets. |

## Exercises and expected answers

| Exercise | Expected |
|---|---|
| Predict: What will sales.shape print? Guess first. | Guess first, then run it to check |
| Write it: Percent of rows with no customer ID, rounded to 1 decimal. | 34.4 |
| Fill in: Rows per country, biggest first. | Your own |
| Predict: How many rows have a quantity of 0 or less now? More or fewer than 1,000? | Guess first, then run it to check |
| Fill in: Quantity above 0, then price above 0. | 102257, 101933 |
| Write it: Real products only: the stock code starts with a digit. | 101459 |
| Fix it: German rows with quantity above 100. | 6 |
| Write it: Remove invoice 541431, with a comment that says why. | (101458, 10) |
| Write it: Total revenue, rounded to 2 decimals. | 1876534.04 |
| Write it: Only the rows from Germany, saved as germany_sales. | (1514, 10) |
| Case: The German warehouse asks: how many orders, and the average revenue per order? | 79 orders, 472.98 per order |
| Extra: Median price, from describe(). | 2.46 |
| Extra: Rows with HEART in the description. | 10816 |

## Cleaning checklist (write the row count after each step)

| Step | Rows left |
|---|---|
| Start | 105,335 |
| Drop duplicates | |

| | |
|---|---|
| Remove cancelled invoices | |
| Quantity above 0 | |
| Price above 0 | |
| Real products only | |
| Remove the giant order | |